

Requirement:

Final report (10 pages maximum, 3000 words, not including figures or references, not including executive summary).

- Summary of findings (one-page executive summary); summary should include a high-level statement of the challenge addressed by the project and the key insights the team generated during the semester. This can include a graphic, but should fit on one page. What's interesting, useful and surprising about your work? Please attach this at the top of the final report PDF when submitting via Canvas.

Boston Public Health Commission

Summary: Forecasting Short-Term Shelter Bed Demand

MIT Analytics Lab Team: Evan Hoch, Jeremie Taranto, Jiao Zhao, Jacob Lebovitz

Problem Statement

This project develops a predictive model to forecast bed demand at Boston Public Health Commission's two shelters. The model aims to forecast two-week bed demand for both shelters to help operational planning and build a decision support interface for BPHC staff to input data and get results.

Challenges

- a. **Predicting Without Future Input:** Future predictors are unavailable when making advance forecasts.
- b. **Too Many Variables and Correlations:** 40+ variables with high multicollinearity and varying levels of missing data.
- c. **Balancing Predictability and Interpretability:** Need both accurate predictions and interpretable insights for planning.
- d. **Predicting Numbers with Confidence Levels:** BPHC requested prediction intervals, but ensemble methods don't produce them directly.
- e. **Inherent Volatility and Data Uncertainty:** High daily variation and risk of missing data or policy changes.

Solutions

- a. Train on lagged relationships (T-14→T), then project forward using current inputs.
- b. Remove correlated variables, apply LASSO regularization, and impute missing values.
- c. Deploy multiple models: linear regression for interpretation, LASSO for balance, ensemble for performance.
- d. Create pseudo-confidence intervals using proportion of predictions within ±c of actuals in cross-validation.
- e. Conduct sensitivity analysis and test with simulated missing data.

Insights

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Boston Public Health Commission

Final Report: Forecasting Short-Term Shelter Bed Demand

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Problem Statement

This project develops a predictive model to forecast bed demand at Boston Public Health Commission's two shelters: 112 Southamptton (men's shelter) and Woods Mullen (women's shelter).

BPHC faces operational challenges due to capacity and demand mismatches. Overflow events occur when demand exceeds available beds, forcing emergency responses and potentially harming living conditions for shelter residents while increasing operational costs. Conversely, overestimating demand leads to wasted labor hours and inefficient resource allocation. BPHC wants to leverage its unique position in data collection to understand the indicators and other factors that contribute to bed demand. By addressing knowledge gaps around the causes of shelter demand and sharing these insights with partner organizations across public health, housing, and social services, BPHC aims to contribute to broader systemic change.

Our primary goal is to develop a predictive model to forecast bed demand over two-week intervals for both shelters. This short-term prediction capability will enable operations teams to make data-informed capacity decisions, reducing both overflow risk and improving living conditions for shelter residents through more reliable bed availability. We will also identify significant variables driving turnout and conduct sensitivity analysis regarding missing data.

Data Overview and Predictive Factors

Literature-Informed Predictors

We approach this problem by examining factors that may be predictive of shelter demand and characterizing their potential impacts. At the macro level, rising homelessness in the United States reflects persistent structural pressures: a chronic shortage of affordable housing, wages that fail to keep pace with rents, and economic shocks such as job loss or unexpected expenses (Austin et al., 2021; Early, 2005; Fitzpatrick, 2005; Rossi & Wright, 1987; Rukmana, 2020). These factors increase the likelihood that individuals lose access

to stable housing and turn to the shelter system for support. At the micro level, immediate conditions strongly influence daily and seasonal patterns in shelter use. Harsh weather (cold temperatures, heavy rain or snow) consistently drives unsheltered individuals to seek indoor refuge (Jadidzadeh & Kneebone, 2015, Jadidzadeh & Kneebone, 2024).

Policy environments surrounding housing and COVID-19 emergency measures may also contribute to shelter bed demand. We included several policy variables: stay-at-home orders, state emergency declarations during COVID-19, President Biden's CHNV and municipal policies such as Ordinance 1373 that displaced encampments, and initiatives like CARE housing that provided temporary rental assistance.

Target Variable: Daily Shelter Occupancy

Our primary outcome variable is daily shelter census, tracked separately for men's and women's shelters from 2015 to 2025. The data reveal substantial daily volatility: mean daily change is 19 individuals (std: 25) for the men's shelter and 8 individuals (std: 11) for the women's shelter. The historical trend shows notable shifts during the COVID-19 period, reflecting both pandemic-related disruptions and policy responses.

Predictor Variables

BPHC provided comprehensive administrative data alongside publicly available indicators, spanning 2015-2025 at daily frequencies.

BPHC supplied detailed administrative data and public indicators from 2015 to 2025 at daily frequencies. Private data cover shelter operations, including daily occupancy and capacity for men's, women's, and other BPHC shelters. Public sources add weather (temperature, precipitation, snowfall), 311 service use by type, monthly unemployment, housing stability measures such as notices to quit, and policy indicators for COVID orders, emergency declarations, federal parole programs, ordinance 1373, care housing efforts, and openings or closings of low threshold shelters. Housing market conditions come from Zillow metrics including total monthly housing payments, observed rent, and new listings.

Predictive Modeling and Results

The goal of our modeling work was to build a forecasting system capable of predicting daily shelter occupancy at BPHC's two major facilities, one for men and one for women. BPHC will use these predictions to determine when to prepare overflow spaces and change staffing levels throughout the seasons.

We will describe our baseline methods to the final model, including how we managed data preparation, what value was created in feature engineering, how we evaluated our model, and the cost of interpretability.

Baseline Model

We began by establishing a baseline against which all more complex models would be evaluated. The most intuitive baseline was an autoregressive rule to use the year before occupancy to predict today's occupancy. We hoped that this baseline would take into account seasonal changes that could account for changes in shelter occupancy. Our baseline model for the men's shelter had an RMSE of 46, and for the women's shelter an RMSE of 24. While this model proved to be a good starting point, we realized that we would have to lower RMSE using other models in order to allow BPHC to use the prediction to improve operations. The large RMSE of the baseline model would not enable better operations due to the high difference between actual vs predicted.

We attributed the large error in this model to its lack of ability to understand policy shocks, weather variation over the year, and daily volatility. Because our data is highly noisy, it is important to further understand the extent to which we will be able to model that noise, and what noise is attributed to true randomness.

This baseline served as a benchmark for assessing improvements delivered by more sophisticated machine-learning and regression models.

Data Preparation and Cleaning

Because our data integrates public, private, and administrative sources spanning 2015–2025, substantial preprocessing was needed to ensure model stability and eliminate outliers.

We identified spurious occupancy spikes using a two-sided deviation rule: if our census variable differed from both of its neighbors by more than 100 people, and the neighboring days remained stable, the point was flagged as a likely data-entry error. Corrections involved replacing these anomalies with the mean of the four nearest neighbors. This procedure adjusted several implausible zeros in men's and women's census records and stabilized the target series.

For our first models using linear regression and tree-based models, columns that contained a substantial amount of NA values (>30%) were still considered in the training feature list. While we tested the performance of the model in excluding them, there was no

significant difference. So, by including these columns, we were able to be assured that our model was robust enough to handle features with missing data. This was an important test for our client as they felt like the final model may have to account for missing data due to delays in third-party data streams.

One key issue we faced was deciding what our target variable would be. In many classical prediction approaches, we were predicting a given y value, in our case, what we thought would be occupancy, based on the data for that feature. However, in this case, when we are predicting 14 days out in the future, we do not have the data, like housing prices or weather, in the future. Therefore, we decided to shift our census values back by 14 days, meaning we were using Day T data to predict turnout on day $T+14$. This key insight allowed us to use machine learning techniques without worrying about missing future data.

Feature Engineering

Shelter occupancy is influenced by social, environmental, and policy dynamics. In making our models, we experimented with feature engineering methods to enhance our results.

To model seasonality without introducing artificial discontinuities, we encoded month and day-of-week using sine-cosine transformations, ensuring December remains adjacent to January, and Sunday remains adjacent to Monday.

Automated filtering eliminated highly correlated predictors (correlation > 0.8), retaining whichever feature exhibited the stronger relationship with the target variable. After processing, approximately 35 predictors remained.

Lasso regression, which is what we used for our linear model, is scale-sensitive, so all continuous features were standardized via Z-score transformation within the training window to ensure fair coefficient penalization. When implemented, the entire uploaded dataset should be used as training to predict the next selected window occupancy.

Model Exploration

To explore a broad modeling space, we evaluated several machine learning techniques:

- Random Forests
- Gradient Boosting Methods (XGBoost)
- LASSO regression with engineered time features
- ARIMA Forecasting using Lagged Census Data

These models capture nonlinearities, interactions, and long-range temporal dependencies to varying degrees. However, interpretability and operational usability were important constraints communicated by BPHC. Thus, although more flexible models such as XGBoost performed competitively, linear models provided clearer insight into key drivers of shelter demand.

Notably, our early experiments included an optimistic “full-information forecasting” model that relied on future X variables, but this approach was not implementable for real operations. This method was used to test hyperparameters because of the speed at which it could produce “testing” accuracy metrics. Other testing methods included taking a smaller subset of data ($t=400$) and predicting the next 14 days for all data points. While this step-forward method is more like the method used in implementation, it requires higher computing time.

Walk-Forward Cross-Validation

To rigorously evaluate the predictive performance of our models under realistic forecasting conditions, we implemented a walk-forward validation procedure. We developed this approach for our time-dependent model, as a traditional train/test data split would not be applicable because we would be “seeing the future.” At each iteration, the procedure trains models on a fixed rolling historical window of 400 consecutive observations. A 14-day gap is then imposed between the end of the training window and the beginning of the test window to prevent information leakage from short-term temporal autocorrelation. The models are evaluated on the subsequent 5 observations, after which the window is advanced forward, and the process repeats.

Within each training window, the chosen modeling method is used to generate optimal predictions for the test window. The different models we experimented with are listed and described in detail below. All predictions are aggregated across walk-forward iterations and aligned with their corresponding true future occupancy values. Model accuracy is quantified using the averaged Root Mean Squared Error across all test predictions.

Below, we explore each tested model, as well as the walk-forward cross-validation metrics for both shelters.

LASSO Regression with Walk-Forward Cross-Validation

Lasso linear regression was used as a model that would yield a good prediction while also offering interpretability in the features. All features were considered and engineered using the above procedure, detailed in the Feature Engineering section.

Model Performance

The final model achieved:

- **RMSE for Census_Men: 34**
- **RMSE for Census_Women: 21**

Tree Ensemble Regression with Walk-Forward Cross-Validation

A tree ensemble of XGBoost and Random Forest was used to balance the bias and variance of XGBoost and Random Forest models. XGBoost has the ability to capture noise in the data, but runs a risk of overfitting on training data. Random Forest predictions are smoother, but fail to capture daily fluctuations. A weighted average (35% XGBoost, 65% Random Forest) of the predictions yielded a balanced model with good performance.

Model Performance

The final model achieved:

- **RMSE for Census_Men: 32.78**
- **RMSE for Census_Women: 15.44**

Ensemble Model Sensitivity Analysis Takeaways

Because public datasets run the risk of lacking information, we tested the impact on these models if we contained missing data. We found that randomly removing 10% of values from a single feature produced no measurable impact on performance. Removing the most recent two weeks of one feature resulted in roughly a 10% performance loss in RMSE. Removing low-importance features, determined through LASSO regression, resulted in less than 5% impact. Removing the most important features (temperature, -14 day occupancy) caused up to a 20% performance drop in RMSE.

BPHC requested that we give an interval of confidence around our predictions. While an ensemble method does not directly lead to a confidence interval, we chose to select a census measure c and find the proportion of predictions within $\pm c$ of the actual value in the walk-forward cross-validation. We did this pseudo-confidence interval for the actual raw point prediction.

- **With 67% confidence our prediction will fall within ____ of the actual turnout using the ensemble model.**
- **With 95% confidence our prediction will fall within ____ of the actual turnout using the ensemble model.**

Relevant graphs and tables are attached in the appendix.

ARIMA Forecasting using Lagged Census

This model uses an autoregressive regression with lagged daily census values to forecast the census 14 days ahead. The model uses:

- Lagged census values at 1, 2, 7, and 14 days
- A one-day difference feature (current census minus census one day ago)
- A seven-day difference feature (current census minus census seven days ago)

Before modelling, extreme spikes in the target series are cleaned using a neighbor mean rule so that isolated outliers do not dominate the fit. The model is estimated with Lasso regression (alpha equal to 0.1), and performance is evaluated using the walk-forward scheme.

Model Performance

The final model achieved:

- **RMSE for Census_Men: 26**
- **RMSE for Census_Women: 13**

With 67% confidence our prediction will fall within 23 of the actual turnout in the men's shelter.

With 95% confidence our prediction will fall within 54.8 of the actual turnout in the men's shelter.

But this model is highly dependent on the quality and availability of the target census input. Because all features are derived directly from the census time series, any missing values or abnormal noise in the target column can significantly degrade forecast accuracy. In cases of severe noise or data gaps, model error can increase drastically.

Since robustness is a key criterion for the BPHC team, we did not want the final production solution to rely on a single-feature-driven model alone.

Final Model

To address this limitation, the final solution uses a dynamic model selection framework that reduces sensitivity to any single data issue. The production system:

- Runs the two best performing models, AR based census model and Ensemble model, on the most recent one month window
- Compares each model's predictions against the observed census values
- Selects the model with the lowest recent error
- Returns only the selected model's forecast for the next two weeks

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This approach ensures that if one model becomes unstable due to noisy, delayed, or missing input data, the system automatically defaults to the more reliable model for that time period.

Limitations and Opportunities for Future Work

Several limitations of the current model offer opportunities for improvement. Future improvements may include probabilistic forecasting, multistep recursive models for X-variable prediction, and integration of additional municipal or policy signals. It is also important to do further exploration of the variance within the original data. The weekly noise should be analyzed more in depth to assess whether there are further variables necessary in the feature set, or if the noise is irreducible and something we should account for through confidence intervals or averaging.

Nonetheless, the current model offers strong predictive performance and should improve operational efficiencies within BPHC.

Design Support Dashboard

We created a decision support interface to offer the operational team a simple, intuitive way to input their shelter data and immediately generate forecasts. The main goal is to make it easy for them to see a variable length outlook and reduce the complexity of daily operational decisions.

The interface includes a few core controls: users can select the type of shelter (men's or women's), choose a forecast horizon ranging from 7 to 35 days, and upload data via CSV. On the right side, it displays visual forecasts with historical trends and predicted values, letting users hover for detailed confidence levels.

By providing a private, secure space for data input and offering clear, visual forecasts, the interface helps the team anticipate shelter demand and plan more confidently. It runs multiple models in parallel, choosing the best results for the current data, and provides a summary of model performance and diagnostics at the bottom.

Recommendations/Conclusion

Findings that help client with capacity planning

Suggestion for future data collection and work

By interpreting variables, we identified LTS site open is a important predictive factors for shelter turnout. After communicating with BPHC, they recommend adding additional data about sweeping movement in the City as a more reliable feature to be considered in the model. This reflects the iterative process of data and modelling, and interpretability of modelling enables communication between data team and operation team, enables the empirical experience informs modelling, and validation of modelling in real-world scenario.

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